

U67 M38813M4 Firmware Static Analysis

Subject: IBM PC110 motherboard U67, marked M38813M4 in the provided schematic. Binary analyzed: M38813E4HP@QFP64.bin.

This report is a static reverse-engineering pass. It is based on the binary, the single-sheet PC110 schematic, and public 740-family documentation. It is not vendor source code and does not claim source-level symbol recovery.

Deliverables generated

Artifact	Path / description
Commented assembly	U67_M38813_commented.asm. Machine-assisted 740-family disassembly with SFR aliases, vector labels, inferred function labels, byte addresses, raw bytes, and comments.
This PDF	Facts, assumptions, vector table, schematic pin mapping, and firmware-behavior notes.
Analysis method	Top-aligned 740-family recursive traversal seeded with vector targets, inspected gates, and direct JSR/JMP targets. Unresolved code/data regions are intentionally left as bytes rather than over-named.

Executive summary

- U67 is the keyboard-controller microcontroller. The schematic labels U67 as M38813/M38813M4 and routes it to keyboard matrix lines, host system data lines, keyboard IRQs, LED/speaker control, and serial/pointing-device style links.
- The binary contains an explicit identity banner at \$C082: "MELPS 740 Series Keyboard Firmware Version 1.1(C) Copyright 1992-1995 RIOS Systems Co.,Ltd.".
- The file is 16255 bytes (0x3F7F). Top alignment places it at \$C081..\$FFFF. This produces a coherent vector table; mapping it at the nominal 16K start \$C080 shifts the vector words and produces nonsensical targets.
- The reset vector under the M3886/3802-family vector model is at \$FFFC and points to \$C208. The code at \$C208 is a short branch into the main reset continuation at \$C1D4.
- Most interrupt vectors point to \$E62C, a three-byte default stub (BRK, BRK, RTI) followed by erased \$FF padding. Two vectors are active-looking: \$FFF0 to \$C0DB and \$FFF6 to \$D088.
- The code heavily uses DBB0, DBB1, DBBSTS0, DBBSTS1, and DBBCON. That strongly matches the documented double-bus interface and the schematic connection to the host data/control bus.
- The firmware is not the PC BIOS. It is embedded keyboard-controller firmware that participates in keyboard matrix scanning, host keyboard-controller traffic, LED/speaker control, and likely pointing-device or auxiliary serial handling.

Input files and hashes

File	Size	SHA-256
M38813E4HP@QFP64.bin	16255 bytes	b29761a2fd39abd9c9419ca73b03beb6c41bc52102e2a7429d1f023f82a2a2b8
PC110-Singlesheet-Schematic.pdf	7259077 bytes	4c1b041feb1c0b7703f2a5c3fbd88e0aaed0157725ed00aebb2dbf98ca702194

Hardware facts from the schematic

The schematic page shows U67 as an M38813M4 device with a 4.00 MHz crystal and a dense mix of keyboard, host-bus, LED, speaker, interrupt, and auxiliary serial nets. The names below are taken from the schematic labels around U67.

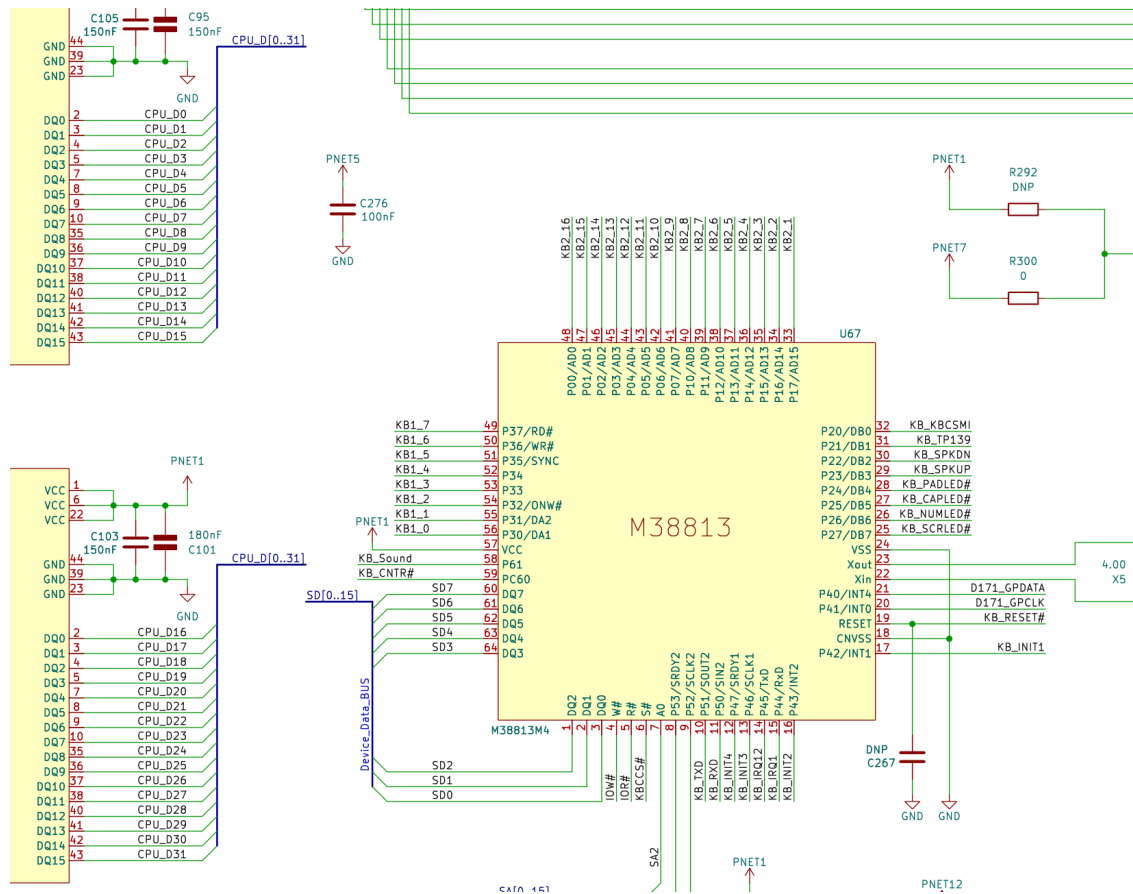


Figure 1. Crop of the PC110 schematic around U67, showing M38813M4, the keyboard matrix nets, host data/control nets, and nearby glue logic.

Area	Observed nets / pins	Firmware significance
Power, mode, reset, clock	VCC pin 57, VSS pin 24, CNVSS pin 18 tied low, RESET pin 19 as KB_RESET#, Xin/Xout pins 22/23 connected to X5 4.00 MHz.	CNVSS low is consistent with normal single-chip operating mode. The first debug checks on hardware should be 4 MHz clock, reset release, and VCC stability.
Keyboard matrix, P0/P1	P00/AD0 through P17/AD15 route to KB2_16..KB2_1 matrix nets.	The binary references P0 and P1 and uses zero-page state heavily. This is the primary matrix-scan side of the controller.
Keyboard matrix / strobes, P3	P30..P37 route to KB1_0..KB1_7 style nets and also overlap RD#/WR#/SYNC alternate functions.	Likely row drive or scan strobes, with alternate bus functions available on the same port group.
Host bus side	DQ0..DQ7 route to SD0..SD7. W#, R#, S#, A0 route to IOW#, IOR#, KBCCS#, and SA2 style nets.	This is the host keyboard-controller style interface. Firmware use of DBB0/DBB1/DBBSTS/DBBCON matches this.
P2 outputs	P20..P27 include KB_KBCSMI, KB_TP139, KB_SPKDN, KB_SPKUP, KB_PADLED#, KB_CAPLED#, KB_NUMLED#, KB_SCRLED#.	Firmware can signal SMI, touchpad/pointing-related state, speaker/keyclick, and keyboard LED outputs.
Serial / auxiliary lines	P50..P53 include KB_INIT3, KB_INIT4, KB_RXD, KB_TXD. P44..P47 include KB_INIT2, KB_IRQ12, KB_IRQ1, D171_P1A2. P40/P41 connect D171_GPDATA/D171_GPCLK.	The SIO1 receive vector and TB_RB/SIO1STS references are consistent with an auxiliary serial channel, likely keyboard/touchpad related.
Host bridge interaction	U35 Pluto nets include KB_CCS, KB_CNTR#, KB_RESET#, KB_IRQ1, KB_IRQ12, PS2_IO, speaker, and LED-related control.	The microcontroller is not isolated. It is integrated into the PC110 chipset and the host-visible keyboard controller path.

Binary layout and memory map

The top-aligned model is the only one that makes the vector table coherent. With a base of \$C081, the vectors resolve to plausible code labels. With a base of \$C080, the same bytes resolve mostly to values such as \$2CE6 and \$2CC0, which do not match real entry points.

Range	Type	Notes
\$C081	Marker byte	First byte is \$E0 before the ASCII identity banner.
\$C082-\$C0DA	ASCII banner	Printable text up to "...Co.,Lt".

Range	Type	Notes
\$C0DB-\$C0DC	Intentional overlap	Bytes \$64 \$2E are both the final two printable banner bytes "d." and the opcode/operand for "tst PCTL1" at the Timer-Y ISR entry.
\$C0DB-\$E62E	Code plus inline tables	Main firmware region. Static traversal recovered 1633 decoded instructions and 3081 traced code bytes. Several blocks remain inline tables or unresolved computed-dispatch data.
\$E62F-\$FFDB	Erased padding	Almost entirely \$FF.
\$FFDC-\$FFFF	Vector table and trailing word	M3886/3802-family vector table through \$FFFC, plus a trailing word at \$FFFE-\$FFFF.

Identity banner

\$C082: MELPS 740 Series Keyboard Firmware Version 1.1(C) Copyright 1992-1995 RIOS Systems Co., Ltd.

The overlap at \$C0DB is unusual but real: ASCII "d." equals bytes \$64 \$2E, which decode as "tst PCTL1" in the 740 instruction set. This lets the firmware keep a full human-readable copyright string while also placing a real vector entry at \$C0DB.

Vector table interpretation

Vector names below follow the public M3886/3802-family vector map used as the nearest available 740-family reference. The reset vector at \$FFFC is independently supported by the reset timing description in the 3802 documentation and by the plausible branch at \$C208. The exact M38813 vector naming should still be checked against the original M38813 manual if it becomes available.

Vector	Family source name	Target	Target label / note
\$FFDC	BRK instruction interrupt	\$FFFF	BRK vector in M3886/3802 map. This image leaves it at \$FFFF.
\$FFDE	A-D converter / key-on wake-up	\$E62C	DEFAULT_ISR_E62C
\$FFE0	INT4	\$E62C	DEFAULT_ISR_E62C
\$FFE2	INT3	\$E62C	DEFAULT_ISR_E62C
\$FFE4	INT2 / I2C	\$E62C	DEFAULT_ISR_E62C
\$FFE6	Serial I/O2 / I2C	\$E62C	DEFAULT_ISR_E62C
\$FFE8	CNTR1 / key-on wake-up	\$E62C	DEFAULT_ISR_E62C
\$FFEA	CNTR0 / SCL/SDA	\$E62C	DEFAULT_ISR_E62C
\$FFEC	Timer 2	\$E62C	DEFAULT_ISR_E62C
\$FFEE	Timer 1	\$E62C	DEFAULT_ISR_E62C
\$FFF0	Timer Y	\$C0DB	TIMER_Y_ISR_C0DB. Real handler with context save.
\$FFF2	Timer X	\$E62C	DEFAULT_ISR_E62C
\$FFF4	Serial I/O1 transmission	\$E62C	DEFAULT_ISR_E62C
\$FFF6	Serial I/O1 reception	\$D088	SIO1_RX_VECTOR_D088. Vector gate plus SIO1 receive body at \$D08A.
\$FFF8	INT1 / output-buffer event	\$E62C	DEFAULT_ISR_E62C
\$FFFA	INT0 / input-buffer event	\$E62C	DEFAULT_ISR_E62C
\$FFFC	RESET	\$C208	RESET_C208. Branches to RESET_CONT_C1D4.
\$FFFE	Trailing reserved word in M3886/3802 map	\$E49E	Trailing reserved word in this map. It contains \$E49E, not reset.

Static disassembly findings

Instruction model

The disassembly uses the Mitsubishi/Renesas 740 instruction set. The public 3802 manual states that the 3802 group uses the standard 740 family instruction set, and the m740dasm project documents the same top-aligned, vector-seeded recursive-traversal method used here. I used an M3886-like SFR table because it contains the same DBB, serial, timer, port, and interrupt-control register names observed in this binary.

Important entry points

Address	Inferred role	Evidence
\$C208	Reset-vector trampoline	Vector \$FFFC points here. It immediately branches to \$C1D4.
\$C1D4	Main reset continuation / polling loop	Enables interrupts, waits on DBBSTS0 and IREQ1 conditions, and calls command/scan handlers.

Address	Inferred role	Evidence
\$C0DB	Timer-Y interrupt handler	Vector \$FFF0 points here. Entry overlaps banner bytes, checks PCTL1, uses ram_69 as a guard, saves A/X/Y and ram_60..ram_63, then calls deeper routines.
\$C20A / \$C211	Host double-bus event gate and body	Checks DBBSTS0 and DBBSTS1, reads DBB0/DBB1, changes DBBCON/DBBSTS bits, saves stack pointer to \$0146/\$0147/\$0148/\$0149/\$014A/\$014B.
\$D088 / \$D08A	Serial I/O1 receive path	Vector \$FFF6 points to a two-byte gate. The body at \$D08A reads SIO1STS and TB_RB and branches to an error path if SIO1STS & \$78 is nonzero.
\$E62C	Default vector stub	Most vectors point here. Bytes decode as BRK, BRK, RTI, followed by \$FF padding.
\$E49E	\$FFFE trailing word target	Begins with RTI. Under the selected vector model, this is not reset. It is preserved as a fact rather than over-interpreted.

Common SFR and RAM references

Address	Symbol	Traced references	Comment
\$60	ram_60	36	Context scratch variable saved in several interrupt paths.
\$63	ram_63	34	Context scratch variable saved in several interrupt paths.
\$64	ram_64	30	State/guard bitfield used by Timer-Y and DBB handlers.
\$08	P4	29	Port P4. On the schematic, P4 pins include interrupt/serial/auxiliary lines.
\$62	ram_62	28	Zero-page RAM or SFR observed in traced code.
\$68	ram_68	25	Zero-page RAM or SFR observed in traced code.
\$61	ram_61	24	Zero-page RAM or SFR observed in traced code.
\$3C	IREQ1	23	Interrupt request register 1.
\$29	DBBSTS0	22	Double-bus status register 0. Heavy host-interface evidence.
\$67	ram_67	19	Zero-page RAM or SFR observed in traced code.
\$00	P0	16	Port P0. Keyboard matrix side.
\$A8	ram_a8	14	Zero-page RAM or SFR observed in traced code.
\$14	ram_14	13	Zero-page RAM or SFR observed in traced code.
\$66	ram_66	12	Zero-page RAM or SFR observed in traced code.
\$2C	DBBSTS1	10	Double-bus status register 1. Host-interface evidence.
\$6B	ram_6b	10	Zero-page RAM or SFR observed in traced code.

Behavioral interpretation

Keyboard scanning. The schematic ties P0/P1/P3 to KB2 and KB1 nets. The firmware references P0/P1 and uses timers and matrix-state RAM. This supports a keyboard matrix scanner role.

Host keyboard-controller interface. The DQ0..DQ7 and R#/W#/S#/A0 pins route to SD0..SD7, IOR#, IOW#, KBCCS#, and SA2. The firmware uses DBB0/DBB1/DBBSTS0/DBBSTS1/DBBCON heavily, which is strong evidence of host double-bus traffic.

Serial or auxiliary device handling. The SIO1 receive vector, SIO1STS error checking, TB_RB reads, and KB_RXD/KB_TXD schematic nets point to an auxiliary serial channel. In the PC110 context, this likely supports keyboard/touchpad or pointing-device side traffic, but static analysis alone does not prove the higher-layer protocol.

LED and speaker control. P2 routes to Caps, Num, Scroll, Pad LED lines plus speaker up/down nets. The firmware likely implements host commands for LEDs and speaker/keyclick behavior.

Nested host-event handling. The DBB event handler saves stack pointers at \$0146..\$014B and switches S using TXS. This is an important design clue: separate paths appear to use different temporary stack frames for host data/control operations.

Repair and debug implications

- For a dead keyboard-controller path, first verify U67 VCC, CNVSS low, KB_RESET# release, and the 4.00 MHz oscillator at X5 on Xin/Xout.
- If the host sees no keyboard controller, probe KB_CCS, IOR#, IOW#, SA2, SD0..SD7, and the U67 DQ pins during host I/O cycles. The firmware expects DBB status/buffer events.
- If matrix keys are dead but the host interface is alive, probe KB1_x and KB2_x activity. P0/P1/P3 are the relevant ports from the schematic.

- If LEDs or speaker functions fail, probe P2 outputs: KB_NUMLED#, KB_CAPLED#, KB_SCRLED#, KB_PADLED#, KB_SPKUP, and KB_SPKDN.
- If pointing or auxiliary input fails, probe KB_RXD, KB_TXD, KB_IRQ1, KB_IRQ12, D171_GPDATA, and D171_GPCLK. The SIO1 receive path reads SIO1STS and TB_RB.

Limitations and unresolved items

- The original M38813 datasheet or mask-ROM listing was not available here. Register names and vector names are mapped from nearby M3886/3802 public references, then checked against the observed binary and schematic.
- The binary is one byte shorter than a nominal 16K \$C080..\$FFFF window. Top alignment is strongly supported by the vector table, but the missing leading byte at \$C080 remains unknown.
- Several regions contain inline tables, computed dispatch, WIT/BRK behavior, or code/data overlap. The assembly deliberately preserves bytes and marks unresolved areas rather than converting everything into named source-level code.
- The exact key matrix map, host command table names, and auxiliary protocol are not fully recovered in this pass. Live tracing or emulator instrumentation would be the next step.

Selected assembly excerpts

Banner overlap and Timer-Y entry

```
$C0DB: 64 2E      tst PCTL1      ; also ASCII bytes "d."
$C0DD: F7 69 01  bbc 7 ram_69,$C0E1
$C0E0: 40         rti
$C0E1: EF 69      seb 7 ram_69
$C0E3: 58         cli
$C0E4: 48         pha
```

Reset vector path

```
$FFFC: 08 C2      .word $C208
$C208: 80 CA      bra $C1D4
$C1D4: 58         cli
$C1D5: A0 01      ldy #$01
$C1D7: 20 BC C1   jsr $C1BC
```

SIO1 receive body

```
$D08A: 48         pha
$D08D: A5 19      lda SIO1STS
$D08F: 29 78      and #$78
$D091: F0 05      beq $D098
$D093: 20 09 E1   jsr $E109
$D0A1: A5 18      lda TB_RB
```

References used

- Provided file: PC110-Singlesheet-Schematic.pdf, page 1. U67 labels and connectivity were taken from the rendered schematic crop and parsed schematic text.
- Acer 390 Series Service Manual, M38813 overview section. It describes M38813M4-XXXHP as an 8-bit single-chip microcomputer for keyboard controller use, with serial I/O, timers, comparator, and double-bus interface.
- Renesas 3802 Group User Manual. Used as a public 740-family reference for instruction set, reset-vector behavior at \$FFFC, interrupt vector ordering, and PROM address mapping notes.
- mnaberez/m740dasm project. Used as an independent reference for top-aligned ROM loading, vector-seeded recursive traversal, and M3886-family SFR/vector naming.